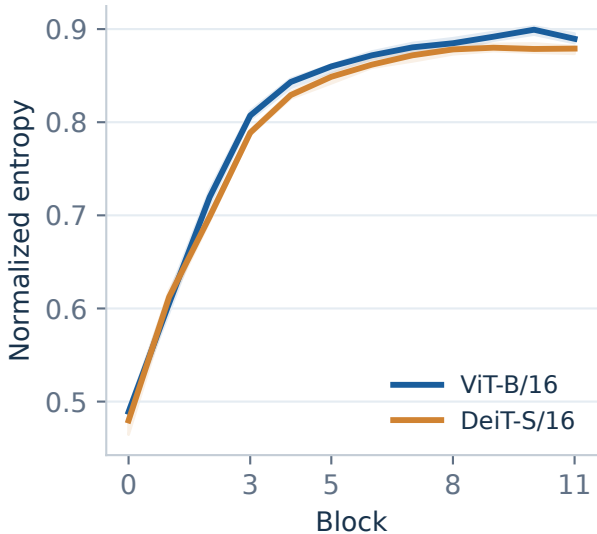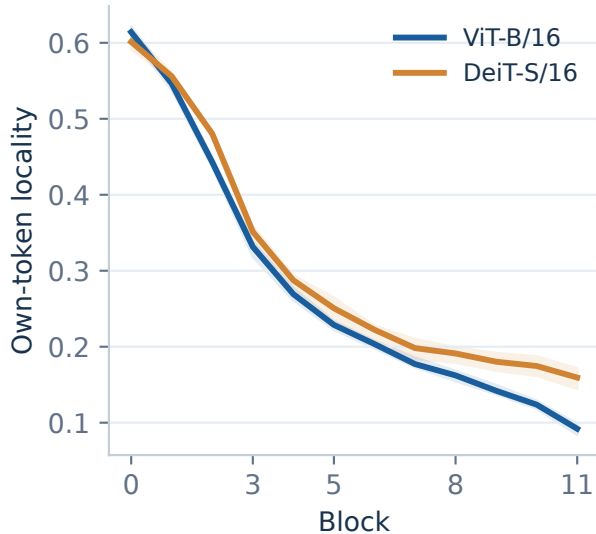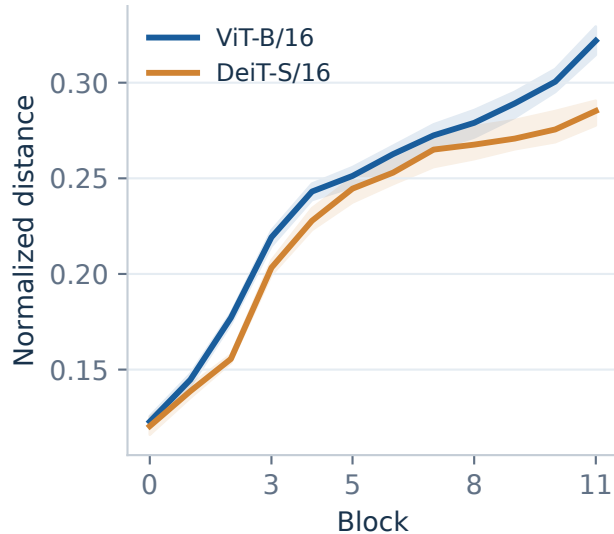Spatial dependence broadens with depth